



AUTOMATED FIELD VALIDATOR

PROPOSED SOLUTION:

Develop an automated system that scans each shipment record for completeness and accuracy. The core goal is to ensure that every required field—such as sender details, destination, parcel contents, and shipment dates—is present and correctly formatted before the parcel is processed.

WORKFLOW:

- Automatically fetch shipment data from a centralized database or API connected to the export business system.
- Run a series of rule-based checks to verify that each mandatory field is filled.
- Flag records with missing or invalid fields in real time.
- Trigger real-time notifications via a dashboard or messaging service for shipments that fail validation.
- Assign compliance scores and rank shipments based on completeness.

TECH STACK

- Frontend: React for the user dashboard, with real-time update capabilities via WebSockets.
- Backend: Node.js/Python for the API and microservices architecture.
- Database: PostgreSQL or MongoDB for shipment data storage.
- DevOps: Docker containers orchestrated with Kubernetes for scalability.

VIABILITY AND FEASIBILITY:

- Viability: The approach uses well-understood rule-based validation which is easy to implement and maintain.
- Feasibility: Given the widespread availability of open-source libraries for data validation, the model is highly feasible. The microservices architecture ensures that the system can scale with growing data volume.



THANK YOU